Dimensionality Reduction and City Ranking Using Principal Component Analysis

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SECTION 1: INTRODUCTION

This research utilizes Principal Component Analysis (PCA) to assess the performance of 329 U.S. cities based on ten socioeconomic and quality-of-life factors: Climate & Terrain, Housing, Health Care & Environment, Crime, Transportation, Education, The Arts, Recreation, and Economics. The data provided reflects multidimensional aspects of city performance. ***Table 1*** gives a quick look at the dataset that was used.

The primary objectives of this analysis are (i) reduce the dimensionality of the dataset while retaining as much of the original variability as possible, (ii) identify latent structures and relationships among the city attributes, and (iii) draw meaningful conclusions regarding relative city performance, including the ranking of cities based on overall performance. Principal Component Analysis is well-suited for this task due to the presence of correlated variables and the need for an interpretable low-dimensional representation of the data as.

Because the ten attributes are measured on different scales, the analysis was conducted using the correlation matrix which effectively standardizes the variables prior to PCA. The analysis was done using Minitab, with Microsoft Excel used for additional computations, visualizations, and result organization. City identifiers(Place ID and Place Name) were excluded from the PCA computations and retained only for labelling and interpretation of results.

The remainder of this report details the data characteristics, the PCA methodology and component retention decision, interpretation of the retained principal components, and conclusions drawn from the analysis of US city performance.

Table 1: Snapshot of Dataset

Place ID	Place Name	Climate & Terrain	Housing	Health Care & Environment	Crime	Transportation	Education	The Arts	Recreation	Economics
1	Abilene, TX	-521	-6200	237	-923	4031	2757	996	1405	7633
2	Akron, OH	-575	-8138	1656	-886	4883	2438	5564	2632	4350
3	Albany, GA	-468	-7339	618	-970	2531	2560	237	859	5250
4	Albany-Schenectady-Troy, NY	-476	-7908	1431	-610	6883	3399	4655	1617	5864
5	Albuquerque, NM	-659	-8393	1853	-1483	6558	3026	4496	2612	5727
6	Alexandria, LA	-520	-5819	640	-727	2444	2972	334	1018	5254
7	Allentown, Bethlehem, PA-NJ	-559	-8288	621	-514	2881	3144	2333	1117	5097
8	Alton, Granite City, IL	-537	-6487	965	-706	4975	2945	1487	1280	5795
9	Altoona, PA	-561	-6191	432	-399	4246	2778	256	1210	4230
10	Amarillo, TX	-609	-6546	669	-1073	4902	2852	1235	1109	6241
11	Anaheim-Santa Ana, CA	-885	-16047	2025	-983	3954	2843	5632	3156	6220
12	Anchorage, AK	-195	-12175	601	-1223	5091	2414	2346	3000	7668
13	Anderson, IN	-530	-5704	580	-878	2865	2469	430	838	3370
14	Anderson, SC	-591	-5725	820	-975	2707	2772	169	613	4262
15	Ann Arbor, MI	-546	-11014	2508	-1067	3433	3346	7559	2288	4579
16	Anniston, AL	-560	-5530	598	-1125	3051	2189	268	1165	4730
17	Appleton-Oshkosh-Neenah, WI	-396	-7877	833	-525	3298	2844	1166	2315	5275
18	Asheville, NC	-694	-6722	1204	-566	5086	2990	1391	1542	5196
19	Athens, GA	-601	-6691	605	-933	1866	2646	3546	1001	5193
20	Atlanta, GA	-696	-8316	3195	-1308	8409	3057	7559	1362	6315

SECTION 2: DATA DESCRIPTION AND CORRELATION STRUCTURE

Table 2: Correlation Matrix

	Climate & Terrain	*Housing*	*Health Care & Environment*	*Crime*	*Transportation*	*Education*	*The Arts*	*Recreation*	*Economics*
Climate & Terrain	1								
Housing	0.386290697	1							
Health Care & Environmen	-0.213302539	-0.453	1						
Crime	0.192387308	0.13422	-0.304659425	1					
Transportation	-0.079119933	-0.2719	0.470317067	-0.28658	1				
Education	-0.064519762	-0.1979	0.490234029	-0.07442	0.335992496	1			
The Arts	-0.226974199	-0.4486	0.865796042	-0.38948	0.464805566	0.37328	1		
Recreation	-0.213509028	-0.4223	0.325396953	-0.34463	0.364716244	0.07781	0.37866	1	
Economics	0.100082741	-0.2694	0.069270737	-0.25999	0.059246753	0.1197	0.07567	0.17353	1
	THRESHOLD	0.3							

Overview of Attribute Correlations

The correlations matrix reveals the presence of multiple moderate to strong correlations among several attributes, indicating that the variables are not independent. This correlation structure suggests that city performance across different dimensions is interconnected rather than isolated, providing an initial justification for the application of PCA.

Positive Correlation Patterns

Strong positive correlations are observed between ***Health Care & Environment*** and ***The Arts***, as well as between ***Health Care & Environment*** and ***Education***. These relationships suggest that cities with strong health and environmental conditions also tend to perform well in cultural and educational dimensions. In addition, ***Transportation*** exhibits moderate positive correlations with ***Education*** and ***The Arts***, highlighting the role of infrastructure in supporting broader quality-of-life outcomes.

Negative Correlation Patterns and Trade-Offs

Several attributes display moderate negative correlations, especially between ***Housing*** and ***Health Care & Environment***, ***Housing*** and ***The Arts***, and ***Crime*** and ***Health Care & Environment***. These negative relationships indicate the presence of trade-offs among certain dimensions of city performance, reinforcing the idea that improvements in one area may be associated with declines in another for some cities.

Attributes with Weaker Correlations

Attributes such as ***Economics*** and ***Climate & Terrain*** show relatively weaker correlations with many other variables. This suggests that these attributes may capture more independent aspects of city characteristics that are not strongly tied to the broader clusters observed in the correlation matrix.

Implications for PCA

Overall, the correlation matrix demonstrates a combination of positively correlated clusters and meaningful negative relationships, providing a strong justification for the use of PCA. The observed correlation patterns indicate that a smaller number of latent components can effectively summarize the variability in the original ten attributes which supports the application of PCA using the correlation matrix.

SECTION 3: PRINCIPAL COMPONENT ANALYSIS AND REDUCED MODEL DEVELOPMENT

Eigenvalues and Variance Explained

Table 3: Eigenvalues and Variance Contribution

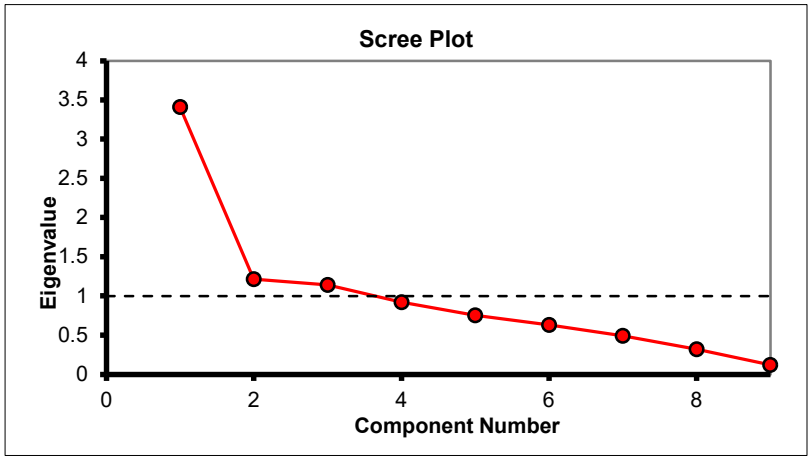
Principal Component	*Eigenvalue*	*Variance Explained (%)*	*Cumulative Variance (%)*
PC1	3.4083	37.9	37.9
PC2	1.2140	13.5	51.4
PC3	1.1415	12.7	64.0
PC4	0.9209	10.2	74.3
PC5	0.7533	8.4	82.6
PC6	0.6306	7.0	89.7
PC7	0.4930	5.5	95.1
PC8	0.3180	3.5	98.7
PC9	0.1204	1.3	100.0

Principal Component Analysis was conducted on the correlation matrix of the ten city performance attributes. The resulting eigenvalues and variance explained indicate that the first principal component has an eigenvalue of 3.41, accounting for 37.9% of the total variance in the dataset. The second and third principal components have eigenvalues of 1.21 and 1.14, explaining an additional 13.5% and 12.7% of the variance, respectively.

The first three principal components together explain about 64.0% of the total variability in the original data. This is a big drop in dimensionality, whilst preserving most of the information from the ten attributes.

Component Retention Criteria

Figure 1: PCA Scree Plot



Three different criteria were used to figure out how many principal components to keep. Kaiser's rule suggests keeping components with eigenvalues greater than one, which is why the first three principal components should be kept. Secondly, the scree plot on ***Figure 1*** has a clear elbow after the third component. This means that components beyond this point add little additional explanatory power. Finally, the first three components explain more than sixty percent of the cumulative variance, which is enough for analyses that use socioeconomic and quality-of-life data.

Selection of the Reduced PCA Model

Based on the eigenvalue magnitudes, scree plot behavior and cumulative variance explained, a reduced PCA model retaining the first three principal components was selected for further interpretation. This reduced model balances dimensionality reduction with interpretability and serves as the basis for subsequent analysis of component meaning and city-level performance comparisons.

SECTION 4: INTERPRETATION OF THE RETAINED PRINCIPAL COMPONENTS

Interpretation focuses on the three primary components preserved in the reduced PCA model. PCA produces a component for each original variable, but only the first three components were kept because of eigenvalues, scree plot behavior, and cumulative variance explained. The following interpretation is based on variables with absolute loadings of at least 0.5, which means they have a strong effect on each component. The table below shows correlations between original variables and principal components.

Table 4: Correlations between original variables Xi and Principal Components Zi

	Loadings	Z1	Z2	Z3	Z4	Z5	Z6	Z7	Z8	Z9
X1	Climate & Terrain	-0.381	0.240	0.737	0.132	0.320	-0.297	-0.059	-0.204	0.000
X2	Housing	-0.658	0.276	0.222	0.491	-0.203	0.112	-0.162	0.346	0.005
X3	Health Care & Environment	0.850	0.330	-0.008	-0.014	-0.090	-0.297	-0.010	0.105	0.249
X4	Crime	-0.519	0.392	-0.198	-0.517	0.455	-0.064	0.013	0.242	-0.020
X5	Transportation	0.648	0.198	0.156	0.291	0.351	0.371	0.410	0.053	-0.001
X6	Education	0.508	0.533	0.245	-0.322	-0.181	0.399	-0.299	-0.106	-0.038
X7	The Arts	0.855	0.215	-0.028	0.097	-0.091	-0.367	0.015	0.115	-0.238
X8	Recreation	0.605	-0.424	-0.054	0.182	0.460	0.071	-0.441	0.085	0.009
X9	Economics	0.250	-0.519	0.649	-0.405	-0.139	0.026	0.105	0.228	0.000

Principal Component 1 (PC1)

The first principal component accounts for 37.9% of the total variance and is characterized by strong positive loadings on ***Health Care & Environment*** (0.850), ***The Arts*** (0.855), ***Transportation*** (0.648), ***Recreation*** (0.605), and ***Education*** (0.508). ***Housing*** (-0.658) and ***Crime*** (-0.519), on the other hand, load strongly in a negative direction.

The size and consistency of these loadings show that PC1 is an effective way to measure how well a city is doing overall and how good(quality) life is there. Cities with high PC1 scores tend to do well in areas like health care, cultural amenities, infrastructure, education, and recreational opportunities. At the same time, they tend to have less crime and better housing conditions. On the other hand, cities with low PC1 scores tend to do poorly in these areas. Because of this, PC1 can be seen as a general index of how well a city is doing, which makes it a good way to rank cities.

Principal Component 2 (PC2)

The second principal component explains 13.5% of the total variance, predominantly influenced by a significant positive loading on ***Education*** (0.533) and a substantial negative loading on ***Economics*** (-0.519). Recreation also has a moderate negative effect on this component.

This pattern indicates that PC2 encapsulates a trade-off dimension between educational achievement and economic or recreational focus. Cities with high PC2 scores tend to care more about education relative to economic strength. On the other hand, cities with low PC2 scores tend to have stronger economies and more opportunities for fun, but comparatively weaker educational outcome. PC2 thus underscores the structural differences in how cities prioritize education in relation to economic and lifestyle considerations.

Principal Component 3 (PC3)

The third principal component accounts for 12.7% of the total variance, characterized by substantial positive loadings on ***Climate & Terrain*** (0.737) and ***Economics*** (0.649). These two variables add a lot more to PC3 than any of the others.

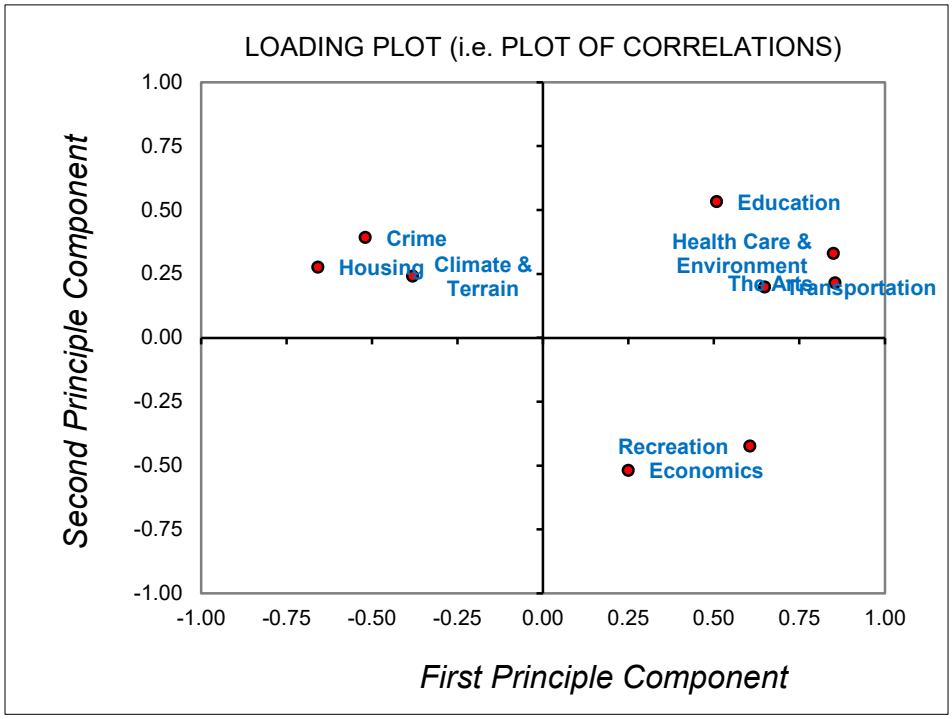
This loading structure shows that PC3 is a climate–economic advantage dimension that divides cities based on their strong economies and good geographic and environmental conditions. Cities with high PC3 scores tend to have better weather and terrain, as well as stronger economies. Cities with low PC3 scores do not have these advantages. PC3 therefore encapsulates a more specific facet of urban performance that is separate from the general quality-of-life aspect indicated by PC1.

The squared loadings further confirm these interpretations, showing that PC1 explains a substantial proportion of the variance in Health Care & Environment and The Arts, while PC3 explains a substantial share of the variance in Climate & Terrain and Economics.

SECTION 5: LOADING PLOT AND ATTRIBUTE GROUPINGS

I looked at a loading plot of the original variables on the first two principal components (PC1 and PC2) to see if the interpretations of the retained principal components were correct (***Figure 2***). This two-dimensional representation gives us useful geometric information about how the city attributes are related because PC1 and PC2 together explain more than half of the total variance.

Figure 2: Loading Plot



Attribute Groupings Along the First Principal Component

The loading plot makes it clear that the attributes are separated along the PC1 axis. This supports the idea that it is a measure of overall city performance and quality of life. Education, the arts, transportation, and health care are all on the positive side of the PC1 axis and are grouped together fairly closely. This grouping shows that these traits are strongly linked to each other and that they all describe cities that do well.

On the other hand, Housing and Crime are on the negative side of the PC1 axis. Their opposing stance regarding quality-of-life attributes suggests adverse correlations with overall city performance, reinforcing the notion that higher-performing cities typically demonstrate reduced crime rates and improved housing conditions.

Trade-Offs Captured by the Second Principal Component

Variation along the PC2 axis shows trade-offs between city features instead of overall performance. Education has a positive load on PC2, while Economics and Recreation have a negative load and are in the lower-right quadrant of the plot. This geometric opposition visually confirms the trade-off structure identified in the interpretation of PC2, where cities that focus on educational outcomes are different from those that focus on economic and recreational features. The separation of these attributes along PC2 shows that this component shows differences in what cities care about, not an overall performance gradient.

Consistency with Interpretations of Components

In general, the loading plot gives strong visual evidence for the ideas in Section 4. In the PCA

space, attributes that were found to be strong contributors to PC1 and PC2 based on numerical loadings also seem to be grouped together or opposed. The agreement between the numerical loading analysis and the geometric representation boosts confidence in the validity and usefulness of the principal components that were kept.

SECTION 6: PRINCIPAL COMPONENT SCORE PLOT ANALYSIS

Interpretation of the Score Plot

Figure 3: Score Plot

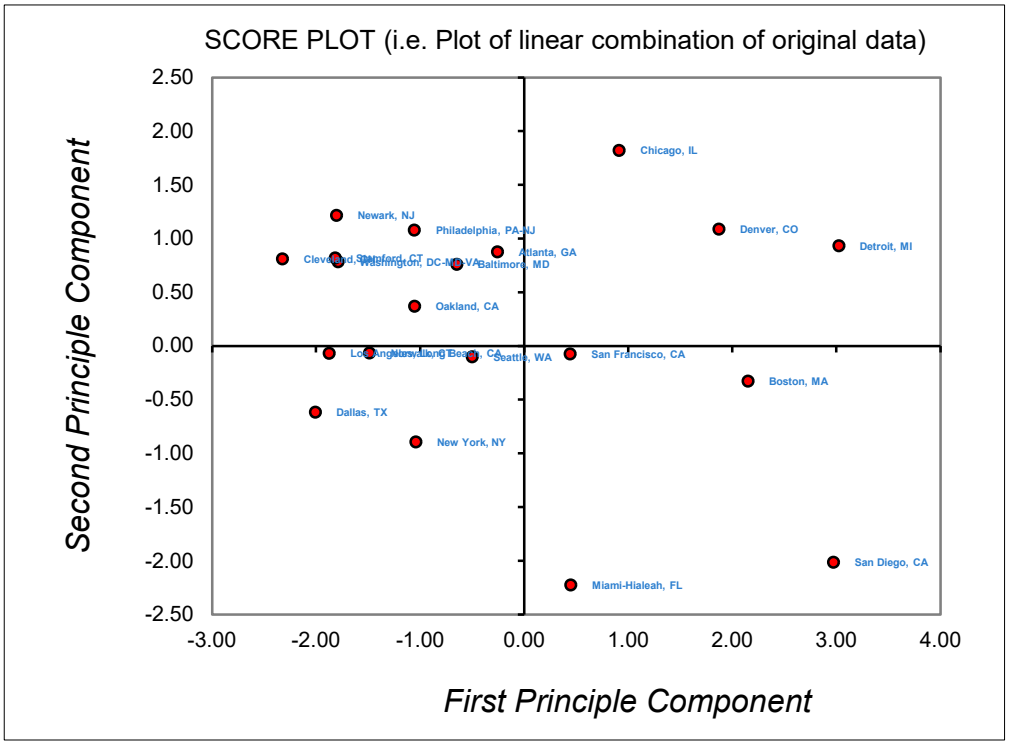


Figure 3 shows the score plot of cities projected onto the first two main components (PC1 and PC2). The position of each point on the graph is based on the city's scores on PC1 (the horizontal axis) and PC2 (the vertical axis). As was said before, PC1 shows how well the city is doing overall, while PC2 shows how city features can be traded off against each other. The arrangement of cities along the PC1 axis indicates big differences in how well they do overall. The figure shows that cities like San Diego, CA; Detroit, MI; Denver, CO; and Boston, MA are on the far-right side. This means that they have strong overall performance across the combined quality-of-life and socioeconomic characteristics measured by PC1. On the other hand, cities like Dallas, TX; New York, NY; and Newark, NJ are on the left side of the plot, which means that this composite metric shows that they are not doing as well overall. The large horizontal dispersion of points shows that cities have different overall performances instead of concentrating around a common profile.

Vertical Separation and Structural Differences (PC2)

Variation along the PC2 axis shows differences in how cities are built that have little to do with how well they do overall. Cities like Chicago, IL and Denver, CO, which are higher on the vertical axis, are more in line with the traits that are linked to high PC2 scores, such as a focus on education. On the other hand, towns like Miami–Hialeah, FL, and San Diego, CA, which are lower on the PC2 axis, are more in line with economic and recreational traits.
This vertical division shows that towns with similar overall performance (identical PC1 scores) may nonetheless have quite different strengths when it comes to education, economics, and recreation.

City Groupings and Outliers

The score plot also reveals clusters of cities with similar performance profiles. Several cities, including Philadelphia, PA; Atlanta, GA; and Baltimore, MD, cluster near the center of the plot, indicating relatively balanced or average performance across the measured attributes.

Additionally, a small number of cities appear as outliers, positioned far from the central cluster. For example, San Diego, CA stands out with a very high PC1 score but a strongly negative PC2 score, while Chicago, IL exhibits a high PC2 score relative to most other cities. These outliers underscore the diversity of city profiles and illustrate how PCA highlights both common patterns and unique cases within the data.

SECTION 7: CITY RANKING AND CONCLUSIONS BASED ON THE FIRST PRINCIPAL COMPONENT

To derive meaningful conclusions from the Principal Component Analysis, cities were ranked based on their first principal component (PC1) scores. PC1 is a strong overall indicator of how well a city is doing because it explains most of the total variance (37.9%) and has favorable effects on important quality-of-life factors like health care and the environment, education, transportation, recreation, and the arts.

The finished PC1 rating demonstrates that big, economically diverse cities are at the top of the list. New York, NY; San Francisco, CA; Los Angeles–Long Beach, CA; Chicago, IL; Boston, MA; Washington, DC–MD–VA; and Philadelphia, PA–NJ are some of the cities with the highest PC1 scores. This means that they do well on many different levels that are all connected. These cities have high composite scores because they have good infrastructure, access to education and healthcare, cultural facilities, and job possibilities.

New York, NY, which has the highest PC1 score, is an extreme positive case since it excels in many areas at once rather than just one. Cities like San Francisco and Boston also get high scores because they have good schools, economies that are driven by new ideas, and good quality of life indices, even though they may have problems with things like housing affordability.

Cities that are in the middle of the PC1 distribution are likely to have strengths that are balanced but not as strong. On the other hand, cities with negative PC1 scores show overall circumstances that are poorer. These cities that are lower on the list are usually smaller metropolitan areas or regions that have fundamental problems with things like access to healthcare, economic diversification, or public infrastructure.

Importantly, the PC1-based ranking should not be seen as a final decision about livability or how well a policy works. Instead, it should be seen as a relative, multivariate comparison. The ranking is based on the cumulative effect of correlated variables, not only performance measurements that are separate. PCA gives us a logical and understandable way to summarize complicated city-level data and find overall performance trends.

In conclusion, ordering cities by the first principal component reveals clear and meaningful differences in the overall city performance. The results demonstrate that PCA is effective in reducing dimensionality, uncovering latent structure, and supporting data driven comparisons amongst metropolitan areas.